Senior Engineers Need Junior Questions

A summary of *Your senior engineers need junior questions more than juniors need answers*
by **Nishant Virmani**, CIO.com, 26 October 2026



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A Quiet Drop in Interruptions

- Juniors now ask an **AI assistant first**, a senior only if that fails
- In one study, **about half** of 131 developers now ask AI what they once asked a teammate
- **6 in 10** said AI is easier to ask: no fear of looking foolish
- Fewer Slack pings, higher velocity: looks like a **win**
- AI also absorbs the trial and error juniors used to learn from

> *Source: UBC & University of Zurich (2026), IEEE TSE study, via Virmani (2026) — see References*

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Two Sides of the Balance Sheet

What juniors lose (widely discussed)

- Less mentoring
- Thinner context
- Shallower feel for production

What seniors lose (rarely discussed)

- Practice explaining the system
- Fresh eyes on confusing code
- A sharp, current mental model

- Senior loss stays hidden because it **looks like focus time**

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What a Naive Question Actually Did

Job	What happened
Passed on tacit memory	Senior tells the story behind a design, e.g. why a service retries three times after a partner's rate-limit incident
Audited for opacity	Newcomer's confusion shows where the code no longer explains itself
Kept the senior sharp	Explaining out loud, under follow-up questions, exposes gaps and outdated assumptions

- All three need a **human on the receiving end**

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The Channel Disappears Without an Alarm

- These questions were never on a calendar or in Jira, so **nobody notices** they stopped
- One sprint: silence feels like focus time
- Several quarters: active command of the architecture **erodes**
- Code review only needs **recognition**; outages need **recall** across services
- Refactoring fell from **25% to under 10%** of changes; duplicated code rose **8×**

> *Source: GitClear (2025), 211M lines of code analysed, as cited in Virmani (2026) — see References*

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Why Velocity Dashboards Hide the Risk

Dashboard shows	Underneath
Shorter pull request cycle times	Fewer people understand why the code is shaped this way
Ticket velocity going up	Juniors close tickets without learning the system
Seniors report fewer context switches	Seniors stop rehearsing the architecture

- Standard metrics don't just miss the erosion, they **read it as progress**

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Three Risks Building Up

- **Single point of failure:** design reasons stay in veterans' heads; shared ownership was really unshared memory
- **Less stable delivery:** every 25% rise in AI adoption linked to a **7.2% drop** in delivery stability
- **Debugging tax:** in a trial, developers learning a new library with AI scored lowest on **debugging**; seniors pick up the fixes later

> *Source: Google DORA (2024); Shen & Tamkin (2026), Anthropic skill formation trial — see References*

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More Documentation Won't Fix It

- First instinct: write more wiki pages, index them into AI search
- That helps **readers** find context
- It doesn't make a veteran **defend trade-offs** or trace failures under questioning
- Banning AI for juniors isn't the answer either
- Optional meetings get cut the moment a deadline hits

> *Documentation captures static answers; only live human friction keeps fluency sharp*

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Rebuild the Channel on Purpose

Moment	Practice
Code review	10-minute live walkthrough for changes touching service boundaries, retries or state machines
Between incidents	Monthly whiteboard walkthrough of a critical workflow, no slides
After an outage	Service owner explains live how the failure spread, no pasted runbook or AI summary
90-day check-in	Can the new hire change a core workflow and explain its failure modes without escalating?

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Counterpoints and Caveats

- **Fewer juniors to ask:** employment of developers aged 22–25 fell about **20%** from its late-2022 peak
- **AI amplifies, it doesn't decide:** DORA 2025 links AI to higher throughput; stability depends on team foundations
- **Usage style matters:** AI users who asked for explanations scored best; heavy delegators scored worst
- **Old channel had costs:** constant interruptions, and one senior answering everything is a single point of failure
- **Core claim untested:** no study yet measures senior fluency decaying

> *Added by SophiArch, not from the original article. Sources: Brynjolfsson et al. (2025); DORA (2025); Shen & Tamkin (2026) — see References*

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Key Takeaways

- Junior questions were **free training for seniors**, not just help for juniors
- Their loss looks like focus time, so **no metric flags it**
- Risk shows up as single points of failure and **slower outage recovery**
- Fix: build **live explanation** into review, walkthroughs, postmortems and onboarding
- Stumbling when explaining your own system isn't lost skill; the **practice was automated away**

> *Adapted from Nishant Virmani, CIO.com (2026) — read the full article via References*

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References — Source and Cited Studies

Topic	Source
Source article	Virmani, N. (2026, October 26). *Your senior engineers need junior questions more than juniors need answers*. CIO, First Person
Help-seeking	University of British Columbia & University of Zurich. (2026). *From Disruptions to Discussions*. IEEE Transactions on Software Engineering
Code quality	GitClear. (2025). *AI Copilot Code Quality: 2025 Research*. GitClear
Delivery stability	Google DORA. (2024). *Accelerate State of DevOps Report 2024*. Google Cloud
Skill formation	Shen, J. H., & Tamkin, A. (2026). *How AI Impacts Skill Formation*. Anthropic. Summary

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References — Further Reading

Topic	Source
AI and delivery, 2025	Google DORA. (2025). *State of AI-assisted Software Development 2025*. Google Cloud
Early-career jobs	Brynjolfsson, E., Chandar, B., & Chen, R. (2025). *Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence*. Stanford Digital Economy Lab
Learning by teaching	Chase, C. C., Chin, D. B., Oppezzo, M. A., & Schwartz, D. L. (2009). Teachable agents and the protégé effect. *Journal of Science Education and Technology*, 18(4), 334–352

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